Spectral sectors and finite-size scaling of fermionic Liouvillian exceptional points

Simultaneous creation of d fermions unfolds a size-$(n+1)$ Liouvillian Jordan block with a fractional exponent $1/(d+1)$. We revisit the uniformly pumped, post-selected quadratic-fermion model of Xu and Yi and resolve the spectrum into finite collective-spin blocks, including the complementary coherence sectors. An exact characteristic-polynomial recurrence clarifies why a truncation linear in the perturbation, although valid at fixed spectral parameter, does not determine all split roots. For fixed d and large n, a semiclassical expansion gives the spectral edge, its first finite-size correction, and the local level spacing. Combining the spin sectors distinguishes unrestricted operator-space decay from parity-even relaxation and from particle-number dynamics. The leading spectral-edge coefficient for $d = 2$ agrees with the known quasi-exactly solvable quartic-potential result. We also specify the trace-nonincreasing parameter range, within which the formal spectral scaling describes a physical conditional evolution.

Introduction.— A high-order exceptional point does not by itself fix the response to a structured perturbation. The perturbation's bandwidth can select fractional powers and several rings of eigenvalues, rather than one ring containing the entire Jordan block [1, 2]. This distinction is relevant to the post-selected fermion dynamics studied by Xu and Yi [3], where uniform d-particle creation perturbs a Liouvillian exceptional point of order $n+1$. Their fractional exponent is correct. What requires refinement is the use of the characteristic polynomial truncated at first order in the perturbation [their Eq. (54), numbering as in arXiv v2]: terms beyond first order need not be negligible on the scale of the split eigenvalues.

Here we give an exact polynomial recurrence and resolve the complementary operator space into the same family of spin matrices. The latter reduction yields exact, sector-dependent formulas for the real-part spectral gaps without diagonalizing a 4^n matrix. A large-spin expansion then supplies explicit finite-size corrections and separates three relaxation scales. We do not claim the fractional exponent or eigenvalue-ring structure as new, and explicitly identify the known $d = 2$ spectral-edge asymptotic [4]. Our purpose is a quantitative spectral and dynamical account of this particular Liouvillian, including the conditions needed for its conditional interpretation.

Model and physical domain.— Let c_j be canonical fermion annihilation operators on n modes, $H = \sum_{ij} h_{ij} c_i^\dagger c_j$ with $h = h^\dagger$, and $\gamma > 0$. Consider

$$\mathcal{L}\rho = -\mathrm{i}[H,\rho] + \gamma \sum_j c_j \rho c_j^\dagger - n\gamma\rho + z\gamma \sum_{|I|=d} c_I^\dagger \rho c_I, \quad (1)$$

where $1 \le d \le n$, $z > 0$, and $c_I^\dagger = c_{i_1}^\dagger \cdots c_{i_d}^\dagger$ for $i_1 < \cdots < i_d$; c_I is its adjoint. Every d-mode subset has the same coefficient. The pump is a recycling term, as in Ref. [3], not a full Lindblad dissipator added to an otherwise unchanged generator. Writing $\hat{N} = \sum_j c_j^\dagger c_j$ gives

$$\mathcal{L}^\dagger \mathbb{1} = \gamma \left[-(n - \hat{N}) + z \binom{n-\hat{N}}{d} \right]. \quad (2)$$

The semigroup is completely positive for $z \ge 0$, and is trace nonincreasing if and only if

$$0 \le z \le \frac{n}{\binom{n}{d}}. \quad (3)$$

Indeed, the omitted jump can have squared amplitude $K = \gamma[n - \hat{N} - z\binom{n-\hat{N}}{d}] \ge 0$; the total retained and omitted jump intensities then equal $n\gamma\mathbb{1}$. Conversely, positivity of K is necessary, and its strongest inequality occurs at $\hat{N} = 0$. Outside Eq. (3) the algebraic spectrum remains defined, but its growing trace is not a post-selection probability for this unchanged generator.

Exact polynomial in the symmetric sector.— Let P_k project onto the k-particle sector without normalization. On $\rho = \sum_{k=0}^n v_k P_k$, the shifted generator $(\mathcal{L} + n\gamma)/\gamma$ is the matrix $N_{n,d}(z)$ with

$$N_{k-1,k} = n - k + 1, \qquad N_{k+d,k} = z\binom{k+d}{d}. \quad (4)$$

All other entries vanish. The physical probability of k particles is proportional to $\binom{n}{k} v_k$, not to v_k alone. With $q = z^{1/(d+1)}$ and $D_{kk} = q^k$,

$$D^{-1} N_{n,d}(z) D = q N_{n,d}(1). \quad (5)$$

Thus the fractional scaling is exact for every nonzero eigenvalue, not merely its leading Puiseux term.

For an explicit polynomial, define $p_\ell(\mu, z)$ as the determinant of the first ℓ rows and columns of $\mu\mathbb{1} - N_{n,d}(z)$, with the full n held fixed. Set $p_0 = 1$ and $p_\ell = \mu^\ell$ for $1 \le \ell \le d$. Then

$$p_\ell = \mu p_{\ell-1} - z w_{\ell-d-1} p_{\ell-d-1}, \quad \ell \ge d+1, \quad (6)$$
$$w_i = \binom{i+d}{d} \frac{(n-i)!}{(n-i-d)!}.$$

A determinant permutation consists of fixed points and nonoverlapping cycles on consecutive intervals of length $d+1$. Consequently,

$$p_{n+1} = \sum_{j=0}^{\lfloor (n+1)/(d+1) \rfloor} (-z)^j W_j \mu^{n+1-(d+1)j}, \qquad W_j > 0, \quad (7)$$

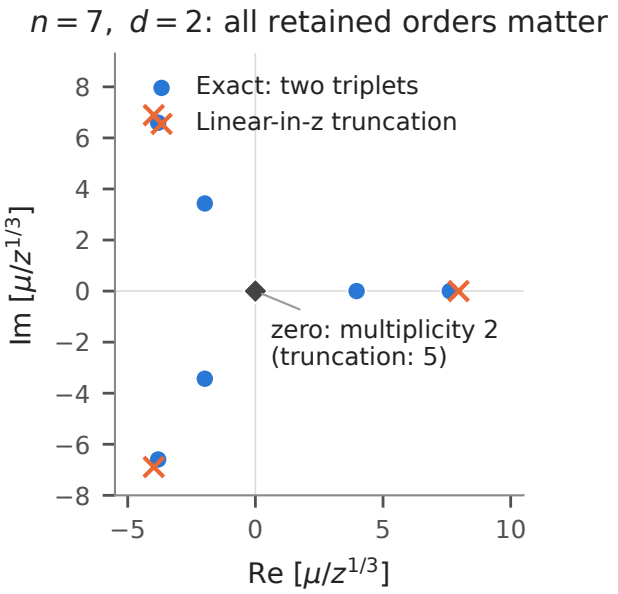


FIG. 1. Exact roots of Eq. (8) (circles) and roots of its linear-in-z truncation (crosses). The axes divide out the exact factor $z^{1/3}$. The difference survives as $z \to 0$, and the residual zero multiplicity is two rather than five.

where $W_0 = 1$ and W_j sums products of the w_i over j disjoint intervals, as in the structured-Jordan analysis of Ref. [1]. The zero eigenvalue has algebraic multiplicity $r = (n+1) \bmod (d+1)$. McMillen's positive double-band theorem [5] further implies that every nonzero root is simple and has the form $\rho_a e^{2\pi i j/(d+1)}$, with distinct positive radii ρ_a. Since the transpose is an unreduced upper-Hessenberg matrix, it is cyclic; when $r > 0$, the zero generalized eigenspace is one Jordan block of size r. This is a special, exactly homogeneous realization of known structured-unfolding results [1, 2, 5].

For $n = 7$, $d = 2$, Eq. (6) gives

$$p_8 = \mu^8 - 504 z \mu^5 + 27504 z^2 \mu^2. \tag{8}$$

There are two cubic triplets and a residual size-two Jordan block (Fig. 1). The linear truncation instead yields one triplet and five zero roots. At fixed nonzero μ, the discarded term is indeed $O(z^2)$. At $\mu = qx$, however, *every* term in Eq. (7) is proportional to q^{n+1}. The truncation is therefore nonuniform on the splitting scale. It is exact when $n+1 < 2(d+1)$; otherwise it does not fix the full root pattern or its coefficients. This limits the inference one can draw from Eq. (54) of Ref. [3], while preserving that work's exponent. Its Fig. 2 uses $n = 4$, where the truncation is complete for $d = 2, 3, 4$; $d = 1$ is independently solved in its Appendix C. Those checks are not contradicted.

Reduction of the complete operator space.— Diagonalize h, with single-particle energies ϵ_j. The loss sum and the sum of all d-particle creation channels are invariant under this change of modes: the latter channels transform by the unitary dth exterior power of the single-particle rotation. This step would fail for generic nonuniform pumping.

In the energy basis, label an operator by disjoint frozen sets A, B, occupied only on the ket or only on the bra, respectively, and a common occupied set C in the remaining $m = n - |A| - |B|$ modes. All vectors $|A \cup C\rangle\langle B \cup C|$ share the Hamiltonian shift $-\mathrm{i}(E_A - E_B)$, with $E_A = \sum_{a \in A} \epsilon_a$. Their fermionic recycling signs are removed by multiplying each vector by

$$g_{A,B}(C) = (-1)^{\sum_{c \in C} |\{f \in A \cup B : f < c\}|}. \tag{9}$$

In this basis the remaining matrix on subsets C is

$$G_{m,d} = J_- + \frac{z}{d!} J_+^d, \qquad J_\pm = \sum_{j=1}^{m} \sigma_j^\pm. \tag{10}$$

One loss removes an arbitrary occupied mode, and one pump adds any d empty modes, with amplitudes 1 and z. The factor $d!$ removes the orderings in J_+^d. The tensor product of m spin-1/2 representations decomposes into spin $s = (m-2j)/2$ with multiplicity $\eta_{m,j} = \binom{m}{j} - \binom{m}{j-1}$, where $\binom{m}{-1} = 0$. Its spin-s matrix is similar to $N_{m-2j,d}(z)$. Therefore the complete spectrum, including multiplicities, is

$$\biguplus_{\substack{A \cap B = \varnothing \\ 0 \le j \le \lfloor m/2 \rfloor}} [-n\gamma - \mathrm{i}(E_A - E_B) + \gamma \operatorname{spec} N_{m-2j,d}(z)]^{\eta_{m,j}}. \tag{11}$$

For an index $M < d$, $N_{M,d}$ is simply nilpotent. Fermionic even-parity operators have $|A| + |B|$ even. Equation (11) is an algebraic block reduction, rather than a closed-form expression for every polynomial root. It applies to arbitrary number-conserving quadratic H, including broken translation symmetry. Jordan-form methods for quadratic open fermions provide broader context [6], but the $d > 1$ recycling term here is not quadratic in fermionic superoperators.

Define $\kappa_{M,d}$ as the Perron root of $N_{M,d}(1)$ and

$$\delta_{n,d} = \kappa_{n,d} - \max_{\nu \in \operatorname{spec} N_{n,d}(1),\, \nu \neq \kappa_{n,d}} \Re\nu. \tag{12}$$

Set $\kappa_{M,d} = 0$ for $M < d$. For $m \ge d$, $G_{m,d}$ is irreducible and nonnegative. Its positive Perron vector is permutation invariant, hence belongs to the maximal-spin sector. Embedding $G_{m,d}$ as a principal submatrix of $G_{m+1,d}$ proves monotonicity of $\kappa_{m,d}$. Thus the leading Liouvillian eigenvalue is $\lambda_* = -n\gamma + \gamma q \kappa_{n,d}$, and the exact real-part separations are

$$\begin{aligned} \Delta_{\mathrm{all}} &= \gamma q \min\{\delta_{n,d}, \kappa_{n,d} - \kappa_{n-1,d}\}, \\ \Delta_{\mathrm{even}} &= \gamma q \min\{\delta_{n,d}, \kappa_{n,d} - \kappa_{n-2,d}\}, \quad n \ge 2. \end{aligned} \tag{13}$$

The minimum cannot generally be dropped: at $n = d = 4$, the symmetric block has fifth-root eigenvalues, including a subleading root with positive real part, whereas $\kappa_{3,4} = 0$.

Large-n edge and level spacing.— Take fixed d and $n \to \infty$, and define

$$\alpha_d = \frac{2d}{d+1}, \quad \beta_d = \frac{d-1}{d+1}, \quad \Omega_d = \sqrt{\frac{d+1}{2}},$$
$$a_d = \left[\frac{2^{-(d-1)}}{(d-1)!}\right]^{1/(d+1)}, \qquad C_d = \frac{a_d}{\alpha_d}. \tag{14}$$

The discrete WKB ansatz $v_k \sim \exp[nS(k/n)]$, at $z = \zeta n^{1-d}$, has the leading symbol

$$\frac{\mu}{n} = h(x,p) = (1-x)e^p + \frac{\zeta x^d}{d!}e^{-dp}, \quad p = S'(x). \tag{15}$$

Minimizing over real p and maximizing the resulting envelope over x gives $x_* = 1/2$ and $\mu/n \sim C_d \zeta^{1/(d+1)}$. In particular,

$$C_d = (d+1)d^{-d/(d+1)}(d!)^{-1/(d+1)}4^{-d/(d+1)}. \tag{16}$$

The saddle coordinate is a spectral construction; it need not equal the mean physical particle filling.

A large-spin expansion supplies the correction and local spacing. Use the orthonormal spin-s representation $\mathsf{N}_s = J_- + zJ_+^d/d!$, similar to Eq. (4), with $s = n/2$. The nonunitary spin rotation $S = e^{\eta J_z}$, where $e^{(d+1)\eta} = zs^{d-1}/(d-1)!$, gives

$$S^{-1}\mathsf{N}_s S = e^{\eta}\left[J_- + \frac{J_+^d}{ds^{d-1}}\right]. \tag{17}$$

Around the $+x$-polarized spin, introduce $[X,P] = \mathrm{i}$, $J_x = s - b^\dagger b$, and $J_y = \sqrt{s}X + O(s^{-1/2})$. The terms linear in X cancel, leaving

$$e^{-\eta}S^{-1}\mathsf{N}_s S = s(1+1/d) + 1 - P^2 - \frac{d+1}{2}X^2 + O(s^{-1/2}). \tag{18}$$

The remainder here is an expansion in the fixed low-excitation sector, not a norm bound on the entire spin space. Its leading term is odd under oscillator parity and has zero diagonal expectation. For fixed oscillator index ℓ, this gives the formal edge expansion

$$\mu_\ell = q\left\{C_d n^{\alpha_d} + a_d[1-(2\ell+1)\Omega_d]n^{\beta_d} + O(n^{\beta_d - 1})\right\}. \tag{19}$$

It predicts the first Perron correction $a_d(1-\Omega_d)n^{\beta_d}$ and, on identifying $\ell = 1$ with the next positive radial root, the spacing

$$\delta_{n,d} \sim 2a_d\Omega_d n^{\beta_d}. \tag{20}$$

McMillen's theorem excludes a competing off-axis edge: those roots have a real-part separation at least $\kappa_{n,d}[1 - \max\{0, \cos(2\pi/(d+1))\}]$, of order n^{α_d}. The identification of the two largest positive roots with the local oscillator levels is the usual edge-localization assumption of this semiclassical expansion; Fig. 2 tests it directly. We do not establish a uniform spectral error bound. Equations (19)–(20) are fixed-d semiclassical asymptotics, not estimates uniform in d/n or in the oscillator index. For $d = 1$ they reduce to the exact spin spectrum $\mu_\ell = q(n - 2\ell)$. For $d = 2$, $N_{n,2}(2)^T$ is the $a = 0$ matrix of Shapiro and Tater [4]; their spectral radius $\frac{3}{4}n^{4/3}[1 + o(1)]$ gives precisely $C_2 = 3/2^{7/3}$. The finite-size correction and the use of the complete Liouvillian sectors are the additional information needed here.

At fixed d, the differences between adjacent spin edges obey $\kappa_{n,d} - \kappa_{n-r,d} \sim r a_d n^{\beta_d}$ for fixed r. Since $2\Omega_d > 2$ for $d > 1$, the edge expansion and Eq. (13) give the asymptotic predictions

$$\Delta_{\mathrm{all}} \sim \gamma q a_d n^{\beta_d}, \qquad \Delta_{\mathrm{even}} \sim 2\gamma q a_d n^{\beta_d}. \tag{21}$$

These differ from the symmetric-block spacing $2\gamma q a_d \Omega_d n^{\beta_d}$. The slowest unrestricted modes are odd-parity coherences with one frozen mode, excluded from physical fermionic density matrices. Even within the physical operator space, a particular observable need not couple to the slowest sector. Total particle number and trace couple only to the permutation-symmetric diagonal sector in the energy basis; their conditional expectation therefore samples the symmetric block, with a generic relaxation exponent $\gamma q \delta_{n,d}$ when the corresponding overlap is nonzero. For $d = 1$, the checks are $\Delta_{\mathrm{all}} = \gamma\sqrt{z}$ and $\Delta_{\mathrm{even}} = \gamma q \delta_{n,1} = 2\gamma\sqrt{z}$.

Scaling within conditional dynamics.— Holding the coefficient of each d-particle channel fixed while increasing n eventually violates Eq. (3) for $d > 1$. A physically admissible large-size scaling is instead $z = \zeta n^{1-d}$ with fixed $0 < \zeta < d!$. In this scaling,

$$\frac{\lambda_*}{\gamma n} = -1 + C_d\zeta^{1/(d+1)} + O(n^{-1}),$$
$$\Delta_{\mathrm{even}} \sim 2\gamma a_d \zeta^{1/(d+1)}, \qquad \Delta_P \sim 2\gamma a_d \Omega_d \zeta^{1/(d+1)}. \tag{22}$$

The gaps can remain finite while the conditional survival weight still decays extensively. For $d > 1$, $C_d(d!)^{1/(d+1)} = (d+1)/(4d)^{d/(d+1)} < 1$, so the leading decay rate stays negative even at the limiting upper bound. A fractional spectral response alone therefore establishes neither improved sensitivity nor a favorable post-selection cost.

Verification and scope.— We checked the polynomial recurrence against exact symbolic characteristic polynomials, including Eq. (8), and the mode rotation and sign gauge against the full superoperator with random complex Hermitian h. Matrix identities, rather than numerical splitting of defective eigenvalues, were used to check the sector reduction. Figure 2 tests both coefficients of the Perron asymptotic and the local spacing. The positive iteration retains every component in logarithmic representation; its Collatz bounds are floating-point convergence diagnostics, not rigorous interval enclosures. Independent 50- and 70-digit spectra agree to better than 10^{-10} in the reported spacings. The accompanying calculation files preserve bounds, precision comparisons, and exact-polynomial checks.

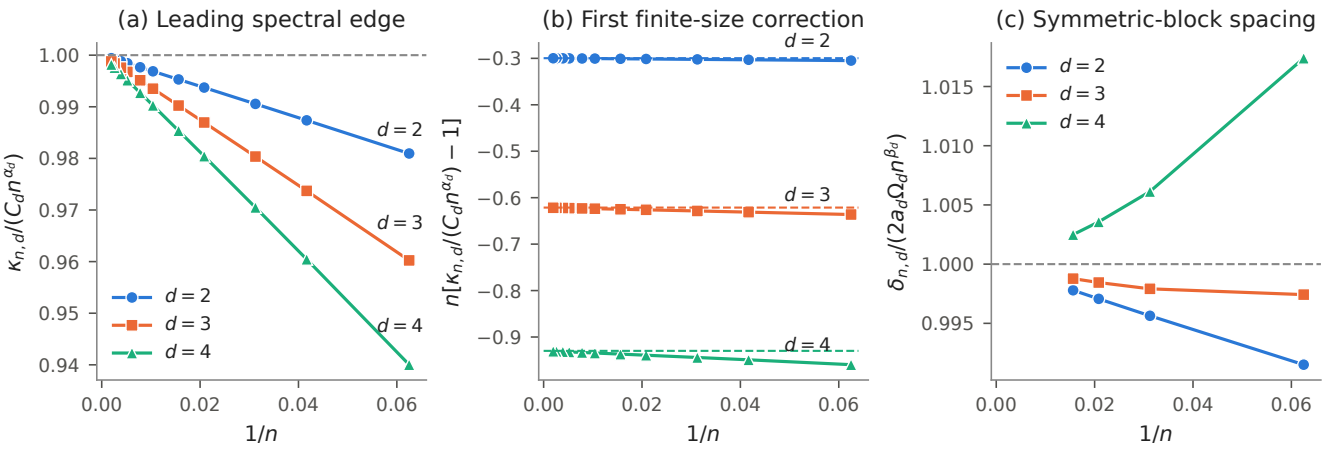


FIG. 2. Parameter-free tests for $d = 2, 3, 4$. (a) Perron root divided by its leading asymptotic value. (b) The scaled residual approaches $\alpha_d(1 - \Omega_d)$ (dashed lines), testing the first correction in Eq. (19). (c) Symmetric-block real-part spacing divided by Eq. (20). Panels (a,b) use log-space positive iteration through $n = 512$ with numerical Collatz bounds of relative width below 2×10^{-12}. Panel (c) uses dense 50- and 70-decimal-digit eigenproblems through $n = 64$. Lines connect data; no coefficient is fitted. The $z = 1$ values shown are rescalable to the physical domain by Eq. (5).

The polynomial correction leaves the established $z^{1/(d+1)}$ exponent intact. The full sector reduction explains which complementary blocks can control relaxation, while the finite-size expansion fixes the coefficients that a fractional exponent alone cannot supply. All results rely on uniform sums over the loss modes and the d-particle creation channels and on number-conserving quadratic H. Nonuniform reservoirs, pairing terms, or additional pump anticommutators require a different spectral analysis.

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